THE REDUCTION OF A FORMAL COMPLETION IS THE REDUCED CLOSED SUBSPACE IT IS TAKEN ALONG

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted spans carry 4 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The reduction of a formal algebraic space and its universal property. Same chapter, Lemma lemma-reduction-formal-algebraic-space, invoked at proof lines 3263-3265. Gives the mapping property of the reduction of the completion.
- The reduced induced closed subspace of a closed subset and its universal property. Properties of Spaces, Definition spaces-properties-definition-reduced-induced-space and Lemma spaces-properties-lemma-reduced-closed-subspace, invoked at proof lines 3265-3269. Gives the mapping property of the reduced closed subspace.
- The definition of the formal completion along a closed subset. Same chapter, the definition of the completion functor, in force in the statement. Supplies the condition that the image of a morphism lies in T.

Source reference. Original lemma 0AIN: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2833–2833. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 047X: sources/stacks-chow/tags.txt, source lines 5524–5524. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 03IQ: sources/stacks-chow/tags.txt, source lines 4612–4612. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $T \subset |X|$ be a closed subset. The reduction $(X_{/T})_{red}$ of the completion $X_{/T}$ of X along T is the reduced induced closed subspace Z of X corresponding to T.*

Proof. It follows from Lemma 0AIN, Properties of Spaces, Definition 047X (which uses Properties of Spaces, Lemma 03IQ to construct Z), and the definition of $X_{/T}$ that Z and $(X_{/T})_{red}$ are reduced algebraic spaces characterized the same mapping property: a morphism $g : Y \to X$ whose source is a reduced algebraic space factors through them if and only if $|Y|$ maps into $T \subset |X|$. □